

Quarter-Car State-Space Model

Corrected derivation | Physical coordinates and road-derivative elimination

1. Force balance

$$\begin{aligned} M_s \ddot{z}_s &= -K_s(z_s - z_u) - C_s(\dot{z}_s - \dot{z}_u) + u \\ M_u \ddot{z}_u &= K_s(z_s - z_u) + C_s(\dot{z}_s - \dot{z}_u) \\ &\quad - K_t(z_u - z_r) - C_t(\dot{z}_u - \dot{z}_r) - u \end{aligned}$$

All displacements use the same upward-positive coordinate and are measured from static equilibrium. Define $s = z_s - z_u$. The Laplace variable is denoted p to avoid confusion with suspension deflection s .

2. Physical state model

$$\begin{aligned} q &= [z_s; v_s; z_u; v_u], \quad v_s = \dot{z}_s, \quad v_u = \dot{z}_u \\ \dot{q} &= A_p q + b_p u + w_p z_r + d_p \dot{z}_r \end{aligned}$$

$$A_p = \begin{bmatrix} 0, & 1, & 0, & 0; \\ -K_s/M_s, & -C_s/M_s, & K_s/M_s, & C_s/M_s; \\ 0, & 0, & 0, & 1; \\ K_s/M_u, & C_s/M_u, & -(K_s+K_t)/M_u, & -(C_s+C_t)/M_u \end{bmatrix}$$

$$b_p = [0; 1/M_s; 0; -1/M_u]$$

$$w_p = [0; 0; 0; K_t/M_u]$$

$$d_p = [0; 0; 0; C_t/M_u]$$

This physical model explicitly needs road velocity. An ideal road step has an impulsive road velocity, causing a wheel-velocity jump. The transformed coordinates below absorb this input derivative.

3. Relative-motion equation

$$\begin{aligned} \ddot{s} &= (1/M_s + 1/M_u) \ddot{u} - (K_s/M_s + K_s/M_u) s \\ &\quad - (C_s/M_s + C_s/M_u) \dot{s} \\ &\quad + (K_t/M_u) (z_s - s - z_r) \\ &\quad + (C_t/M_u) (\dot{z}_s - \dot{s} - \dot{z}_r) \end{aligned}$$

Transformed coordinates

The fourth state is an auxiliary coordinate, not a physical velocity.

4. Define the transformed state

```
alpha = Cs/Ms + Cs/Mu + Ct/Mu
eta = s_dot - (Ct/Mu)*zs + alpha*s + (Ct/Mu)*zr
x = [zs; zs_dot; s; eta]
```

Differentiate eta and substitute the relative-motion equation. The terms in zs_dot , s_dot and zr_dot cancel exactly:

```
eta_dot = (Kt/Mu)*zs - (Ks/Ms+Ks/Mu+Kt/Mu)*s
          + (1/Ms+1/Mu)*u - (Kt/Mu)*zr
```

```
s_dot = (Ct/Mu)*zs - alpha*s + eta - (Ct/Mu)*zr
```

Substitute this expression for s_dot into the body force balance. Together with $zs_dot = x2$, this gives $x_dot = A*x + Bu*u + Bw*zr$ and $s = C*x$:

```
alpha = Cs/Ms + Cs/Mu + Ct/Mu
```

```
A = [ 0,          1,  0,          0;
      -Cs*Ct/(Ms*Mu), 0,  Cs*alpha/Ms-Ks/Ms, -Cs/Ms;
      Ct/Mu,        0, -alpha,        1;
      Kt/Mu,        0, -(Ks/Ms+Ks/Mu+Kt/Mu), 0 ]
```

```
Bu = [0; 1/Ms; 0; 1/Ms+1/Mu]
```

```
Bw = [0; Cs*Ct/(Ms*Mu); -Ct/Mu; -Kt/Mu]
```

```
C = [0, 0, 1, 0]
```

5. Recover physical motion

```
zs = x1;  zs_dot = x2;  zu = x1-x3
s_dot = A(3,:)*x + Bw(3)*zr
zu_dot = zs_dot - s_dot
zs_ddot = (-Ks*s - Cs*s_dot + u)/Ms
```

For a road jump dzr , the transformed x remains continuous. The recovered wheel velocity jumps by $(Ct/Mu)*dzr$. This is a consequence of the ideal tire damper and ideal discontinuous road, not a state-integration error.

Transfer functions and checks

Consistent channels, output sign, and integral augmentation

6. Transfer functions

Taking Laplace transforms with zero initial conditions gives a 2-by-2 dynamic-stiffness system. Eliminating the body and wheel displacements yields:

$$D(p) = (M_s p^2 + C_s p + K_s) * (M_u p^2 + (C_s + C_t) p + K_s + K_t) - (C_s p + K_s)^2$$

$$G_u(p) = S(p)/U(p) = ((M_s + M_u) p^2 + C_t p + K_t) / D(p)$$

$$G_w(p) = S(p)/Z_r(p) = -M_s p^2 (C_t p + K_t) / D(p)$$

The output is always $S = Z_s - Z_u$. G_u has force input; G_w has road displacement input. Reversing the output to $Z_u - Z_s$ changes the sign of both transfer functions.

7. Integral augmentation

```
xi_dot = C*x
xa = [x; xi]
Aa = [A, zeros(4,1); C, 0]
Bua = [Bu; 0]; Bwa = [Bw; 0]
```

Use B_{ua} for controller design and retain B_{wa} as the disturbance channel. The integral of suspension deflection is distinct from η . In passive mode the unused integral state has a zero eigenvalue; assess passive mechanical stability using A .

8. Independent consistency checks

```
x = T*q + h*zr
T = [1, 0, 0, 0;
     0, 1, 0, 0;
     1, 0, -1, 0;
     alpha-Ct/Mu, 1, -alpha, -1]
h = [0; 0; 0; Ct/Mu]
```

```
A = T*Ap/T; Bu = T*bp
Bw = T*wp - A*h; T*dp + h = 0
```

The companion validation checks these identities, both transfer functions across a frequency grid, passive stability, and controllability of both the original and integral-augmented actuator plants. See `VALIDATION.md` for execution details.